

UNIT 1.2

Emotional Design – Three levels of Design – Visceral, Behavioral and Reflective design; designs with personality – machines that senses emotions and induce emotions- Robots, personality products, products for games, fun, people and places; Simulation – dimensional or mathematical, virtual simulation, physical simulation, scale down models;

1.2.1 EMOTIONAL DESIGN

- Emotional design refers to the practice of creating products, systems, or interfaces that evoke specific emotions and create positive user experiences. It goes beyond functionality and usability by considering how a design makes users feel. Emotional design can **enhance user satisfaction, increase engagement, and foster a deeper connection with the product.**
- Emotions are inseparable from and a necessary part of cognition.
- Everything we do, everything we think is tinged with emotion, much of it sub conscious.
- In turn, our emotions change the way we think, and serve as constant guides to appropriate behavior steering us away from the bad, guiding us towards the good.

Emotional Design is a design approach that aims to create products that not only perform their intended functions but also evoke specific emotions, enhancing user experience and satisfaction.

Emotional design plays a critical role in user engagement, helping to build a strong emotional connection between the product and its users. This connection often leads to higher customer loyalty, repeat use, and positive word-of-mouth.

Influence on interaction:

Emotions play a critical role in how users interact with a product. Positive emotions like joy or satisfaction can enhance the user experience, while negative emotions like frustration or confusion can detract from it.

Emotional Triggers:

Identifying and utilizing triggers (such as color, tone, or imagery) that evoke desired emotions is key to creating a compelling user experience.

Product loyalty:

Emotional engagement can transform casual users into loyal advocates. When users have positive emotional experiences, they are more likely to return to the product and recommend it to others.

Psychological aspects:

Explore how psychological principles like the “peak-end rule” (where people judge an experience largely based on how they felt at its peak and at its end) can be applied in emotional design.

Principles Of Emotional Design

Emotional Design

Creating enduring, delightful, and engaging product experiences



Don Norman's emotional design explores the connection between a user's emotional response and their interaction with products

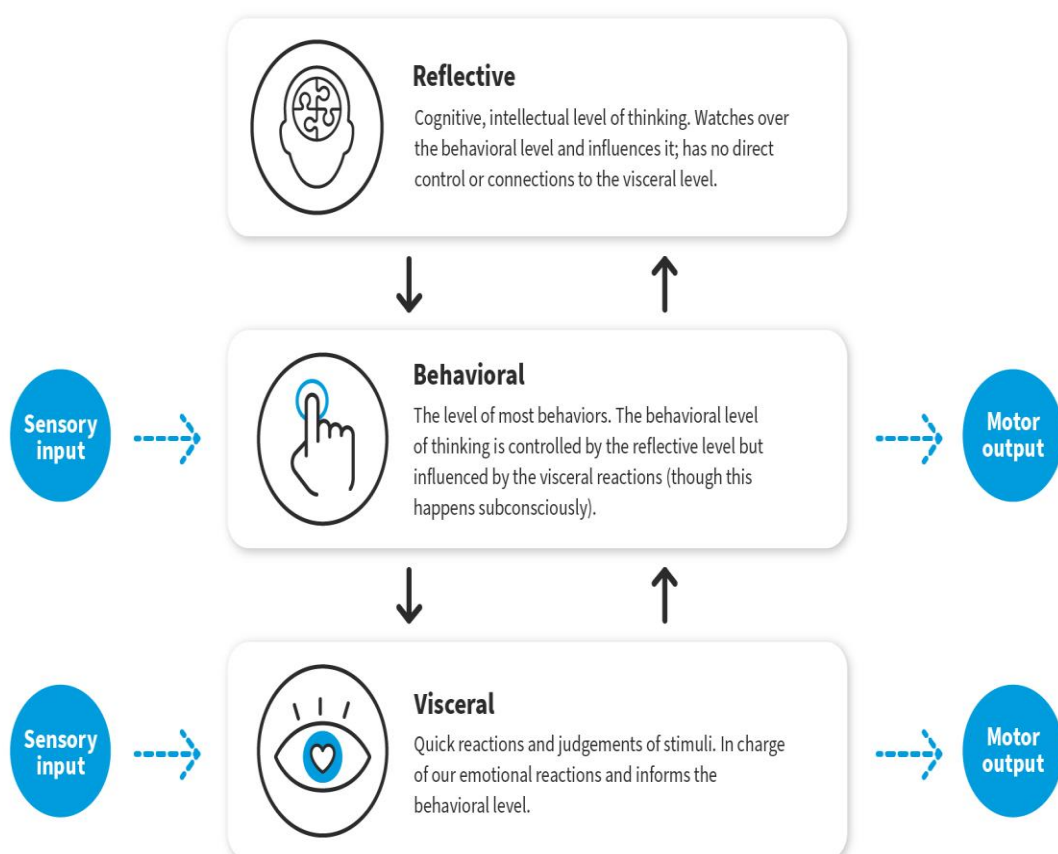
Emotional design focuses not just on how products make people feel, influencing their perception and attachment to the product.

Norman outlines three key levels of emotional design:

- Visceral Design
- Behavioral Design
- Reflective Design

Don Norman's 3 Levels of Design

Source: *Emotional Design: Why We Love (or Hate) Everyday Things* by Don Norman



KEY ASPECTS OF EMOTIONAL DESIGN

1. ***Aesthetics***: Visual appeal and attractiveness can elicit positive emotional responses.
2. ***User Experience***: Ensuring the product is enjoyable and intuitive to use.
3. ***Personality***: Giving the product a distinct character or personality that users can relate to.
4. ***Emotional Engagement***: Creating features or interactions that elicit joy, surprise, or other emotions.

Emotional design is often used in various fields such as consumer electronics, automotive design, software interfaces, and branding to create memorable and impactful user experiences.

1.2.2 VISCERAL DESIGN

- Visceral design is what nature does. We humans evolved to coexist in the environment of humans, animals, plants, landscapes, weather, and other natural phenomena. As a result, we are exquisitely tuned to receive powerful emotional signals from the environment that get interpreted automatically at the visceral level.
- Concerns the initial impact and appearance of a product, focusing on sensory experiences such as aesthetics, texture, and sound.
- Visceral design is about the look, feel, and sound of a product. It creates an initial emotional response based on sensory input.
- Visceral design focuses on the immediate, subconscious reactions users have when they first encounter a product. It is all about the appearance, feel, and overall sensory impact.

- Includes aspects like color, shape, texture, and overall aesthetic appeal. Visceral design aims to create a strong first impression that captures the user's attention and elicits an emotional response.



KEY ELEMENTS OF VISCERAL DESIGN

- Visual appeal: Shape, color, form
- Sensory experience: Texture, sound, and even smell
- Immediate perception: The first impression that captures attention.

Example: The sleek and minimalist design of the iPhone, which immediately attracts users with its visual appeal and high-quality materials.

The smooth curves and high-quality materials of luxury cars like those from Porsche or Mercedes-Benz evoke a sense of sophistication and exclusivity.

1.2.3 BEHAVIORAL DESIGN

- Relates to the functionality and usability of a product, ensuring it meets the practical needs of the user effectively.
- Behavioral design is all about use. Appearance doesn't really matter. Rationale doesn't matter Performance does.
- Behavioral design ensures a product works well and is easy to use. It focuses on how the user interacts with the product.
- Effective behavioral design leads to high user satisfaction as it makes interactions effortless and satisfying, encouraging repeated use.
- What matters here are four components of good behavioral design:
 - a. Function
 - b. Understandability
 - c. Usability
 - d. Physical feel.

Key Elements:

Functionality: Ensuring the product works efficiently and reliably.

Usability: Creating a seamless user experience where the design helps users achieve their goals with minimal effort.

Feedback: Providing clear feedback to users on their actions, helping them feel in control.

Good behavioral design should be human – centered, focusing upon understanding and satisfying the needs of the people who actually, use the product.

Apple products are renowned for their ease of use, with intuitive interfaces that make even complex tasks simple and enjoyable.

Example:

Tesla cars offer a highly intuitive interface, making advanced technology accessible and enhancing the driving experience. Which provides a seamless and user-friendly driving experience through advanced technology and easy-to-use controls.

Google Search Engine: Its simple, clutter-free interface makes searching the web easy and quick, which has contributed to its widespread adoption.

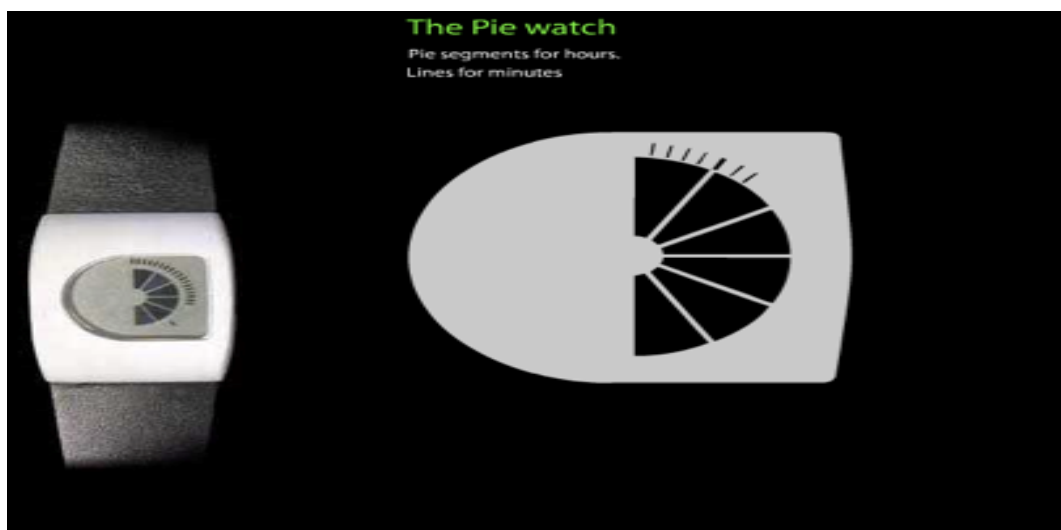
Nest Thermostat: Designed to be easy to control with a simple twist of the dial, making temperature management intuitive.

1.2.4 REFLECTIVE DESIGN

- Reflective design covers a lot of territory. It is all about message, about culture, and about the meaning of a product or its use.
- The personal significance and cultural meaning of a product, shaping the user's identity and emotional attachment.
- Reflective design considers the meaning and personal attachment a user has with a product, influencing how they perceive themselves.
- Reflective design pertains to the meaning and personal significance a product holds for the user. It's the level where users think back on their experiences and form deeper connections with the product.

Key Elements:

- **Cultural Significance:** How a product resonates with a user's cultural background and values.
- **Self-Image:** Products that contribute to the user's self-image or status can create a powerful emotional bond.
- **Memories and Associations:** Products that users associate with positive memories or milestones in life often gain sentimental value.



Reflective design through cleverness

- Designs which are dominated only by function – Reflective Design
- Luxury Watches: Often seen as symbols of success and achievement, they carry significant reflective value.
- A Rolex watch symbolizes luxury and success, providing emotional value beyond its functional purpose of telling time.
- Prestige, perceived rarity, and exclusiveness work at the reflective level.

Example:

- Bose headphones are associated with premium sound quality and noise-cancellation technology.
- Many users feel that owning a pair reflects a sophisticated appreciation for audio experiences.

What will my friends think when they see me wearing this watch?

The reflective level mediates the effects of the behavioral level users may well put up with difficulties and shortcomings in the usability of the smartwatch because they believe they will gain other, non-functional benefits from it.

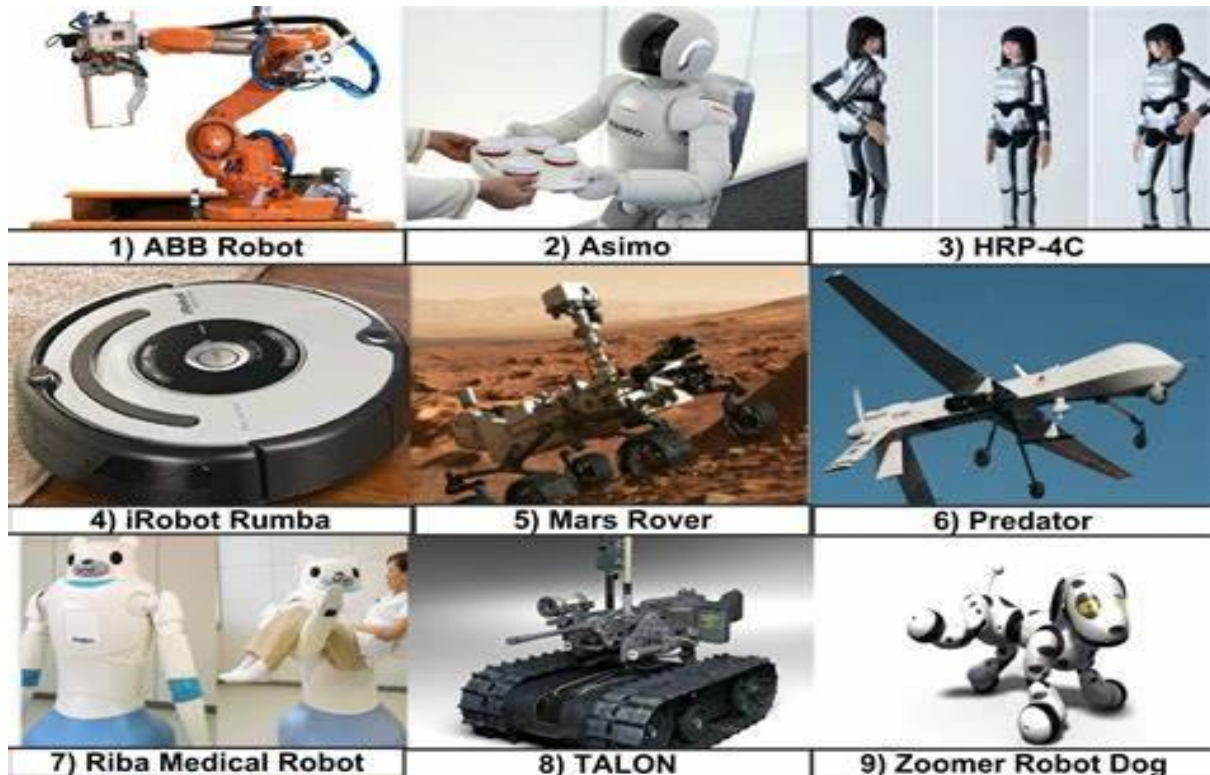
1.2.5 DESIGNS WITH PERSONALITY – Robots with AI

Designing robots with personality involves integrating AI technologies with thoughtful design elements to create machines that can interact with humans in engaging and meaningful ways.

Key aspects to design with personality are

- Aesthetics
- Expressive features
- Movement Gestures

Designs with personality involve creating objects, spaces, or interfaces that reflect distinct character traits, emotions, or narratives. This can be achieved through several elements like color, form, texture, and interaction, making the design feel alive or relatable.



Color Choices:

Colors evoke emotions and can significantly influence how a design is perceived. Bright, bold colors (e.g., neon, saturated hues) can suggest energy and excitement, while muted tones (e.g., pastels or earth tones) can evoke calmness, warmth, or elegance.

Typography:

Fonts with distinct characteristics, such as handwritten or playful script fonts, often give designs a more personable, whimsical feel. In contrast, sleek and minimalist fonts like sans-serif can communicate professionalism or modernity.

Illustrations and Icons:

Custom illustrations or hand-drawn elements add a human touch to a design, making it feel more relatable or friendly. For example, using doodles or quirky sketches can make the design playful.

1.2.6 MACHINES THAT SENSES EMOTIONS AND INDUCE EMOTIONS

Machines that sense and induce emotions, which seeks to create systems that can recognize, interpret, and simulate human emotions.

Machines are very good at analyzing large amounts of data, they can listen to voice inflections and start to recognize when those inflections correlate with stress or anger. Machines can analyze images and pick up subtleties in micro-expressions on humans' faces that might happen even too fast for a person to recognize.

Cogito's voice-analytics software is based on years of human behavior research to identify voice patterns

1.2.6.1 SENSING EMOTION:

A machine would need to detect emotional states through physiological and behavioral cues. Here are some potential methods:

Facial Recognition and Expression Analysis: AI-driven cameras can capture and analyze facial expressions to infer emotions (e.g., happiness, sadness, anger).

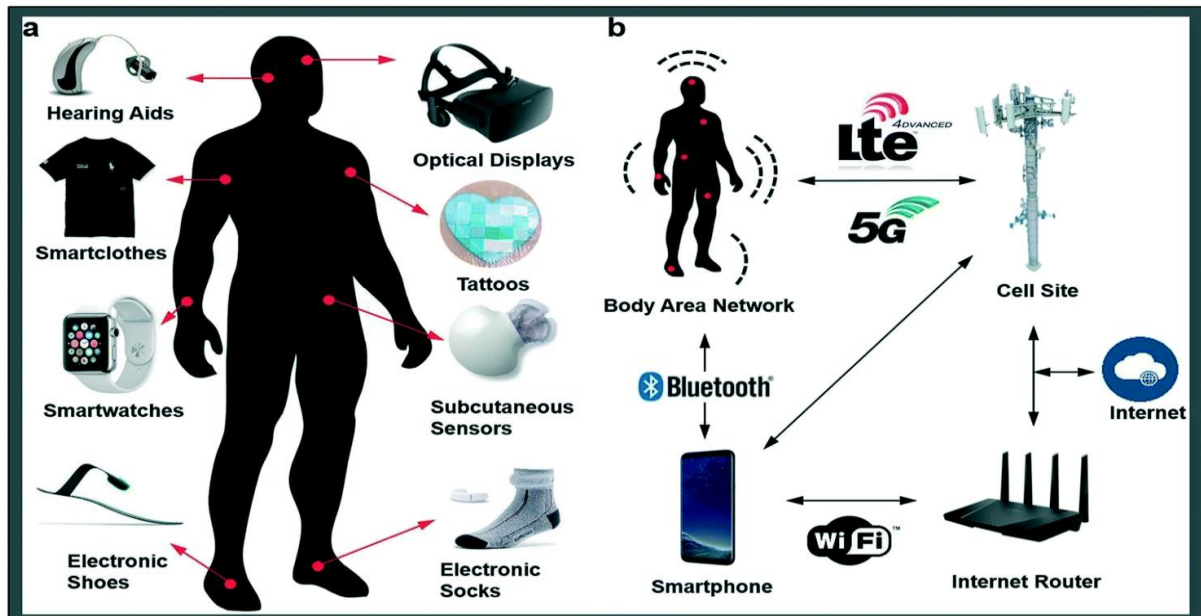
Speech Analysis: Tone, pitch, and speed of speech can provide cues about a person's emotional state.

Physiological Sensors: Devices like wearable wristbands or smartwatches can monitor heart rate, skin conductivity (GSR), and temperature to detect emotional arousal (stress, excitement).

EEG or Brainwave Sensors: Advanced systems could measure brainwave patterns to detect more specific emotional states like calmness or anxiety.

Eye Tracking: Pupil dilation and blinking patterns can also be indicators of emotional states.

Body Language Detection: Cameras and motion sensors could capture posture, movement speed, and other non-verbal cues.



1.2.6.2 INDUCING EMOTION:

To induce emotions in users, machines can interact through various stimuli:

Visual and Auditory Stimuli: Dynamic lighting, colors, and sounds can be used to evoke particular emotions (e.g., calming visuals and soothing music to induce relaxation).

Haptic Feedback: Vibrations or touch-based stimuli can be designed to create comforting or energizing sensations.

Virtual Reality (VR): Immersive VR experiences could be tailored to evoke specific emotional responses by placing the user in certain scenarios.

Neurofeedback: Devices using transcranial magnetic stimulation (TMS) or other neurostimulation methods could theoretically alter brain activity to induce certain emotions.

Scent and Aroma Therapy: Aromas have a profound effect on emotional states and could be used to evoke feelings of calm, happiness, or energy.

Another emotion AI-driven technology for mental health is a **wearable device** developed at the MIT Media Lab that monitors a person's heartbeat to tell whether they are experiencing something like stress, pain, or frustration. The

monitor then releases a scent to help the wearer adjust to the negative emotion they're having at that moment.

Assistive services

Some people with autism find it challenging to communicate emotionally. That's where emotion AI can be a sort of "assistive technology," Hernandez said. Wearable monitors can pick up on subtleties in facial expressions or body language in someone with autism (like an elevated pulse rate) that others might not be able to see.

Hernandez said there are also "communicative prostheses" that help autistic people learn how to read other's facial expressions.

One example is a game in which the person uses the camera on a tablet to identify "smiley" or "frowny" faces on the people around them.

"That is a way for them to engage with other people and also learn how facial expressions work," video technology that measures moods in "smiley" or "frowny" faces could work for customer feedback in crowded theme parks or hospital waiting rooms, or could be used to provide anonymous feedback to upper management in a large office.

Transcranial Magnetic Stimulation (TMS) is a non-invasive procedure that uses magnetic fields to stimulate nerve cells in the brain. It is primarily used to treat depression, especially for patients who have not responded to medication. TMS is also being researched for other conditions such as anxiety and post-traumatic stress disorder (PTSD).

1.2.7 ROBOTS, PERSONALITY PRODUCTS, PRODUCTS FOR GAMES, FUN, PEOPLE AND PLACES

Buddy:

A companion robot that offers various services, including smart home integration and emotional support. Buddy is designed to be friendly and approachable, helping to reduce loneliness. Buddy is designed to create a supportive and engaging environment that helps individuals with autism develop social skills, emotional awareness, and a sense of companionship.

Buddy is designed to support individuals with autism

1. Social interaction
2. Emotional support
3. Routine and structure
4. Parental support
5. Interactive learning
6. Calming presence

Aibo: Sony's robotic dog Aibo mimics real pet behavior and develops its personality based on interactions with its owner. It can express emotions, recognize commands, and create a unique bond with its user.

LEGO Boost: This building kit allows users to create and program their own robots. It combines construction with gaming, offering a range of challenges and interactive experiences through its accompanying app.



Anki Cozmo:

A small robot that combines play with personality. Cozmo can play games, solve puzzles, and interact with users, learning from their interactions over time.

Tesla Optimus is designed to perform tasks that are repetitive, dangerous, or boring for humans. The goal is to assist in various industries, including manufacturing, logistics, and home environments.

Optimus stands about 5 feet 8 inches tall and is designed to mimic human movements and dexterity, making it capable of handling everyday tasks.

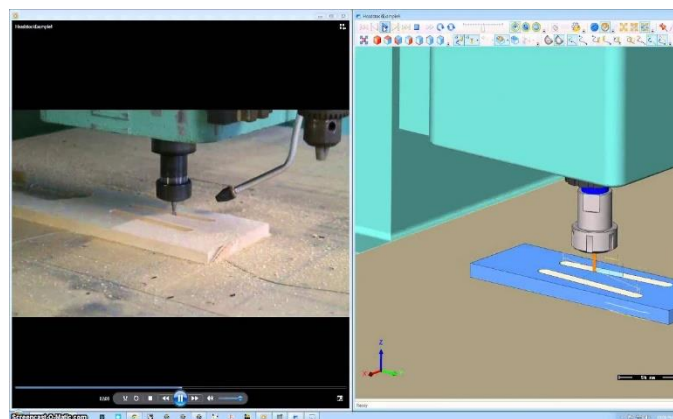
The robot utilizes Tesla's advanced AI and machine learning technologies, allowing it to navigate environments, recognize objects, and interact with people.

1.2.8 SIMULATION

- **Simulation** is the process of creating a model or representation of a real-world system or processes. It provides a way to study and analyze complex phenomena under various conditions.
- Simulations can be used to test hypotheses, analyzing behaviors, predicting outcomes, performing risk assessments, and design new systems.

1.2.8.1-DIMENSIONAL SIMULATION OR MATHEMATICAL SIMULATION

Dimensional simulation refers to the use of technology to create immersive, multi-dimensional experiences that replicate real-world environments or scenarios.



- Dimensional simulation refers to the use of mathematical models to represent and analyze the behavior of real-world systems across different dimensions.
- Dimensional simulation is the process of creating a mathematical model to represent a physical system. This model includes various dimensions, such as space, time, and other relevant variables, to accurately simulate how the system behaves.

- These simulations often leverage physical models to explore how various factors interact in a controlled environment, allowing for experimentation and prediction of outcomes without direct interference in the real world.
- Building simulation is a way to test how elements of building design will perform under real-world conditions.
- This is done using a mathematical model that recreates conditions in a virtual world
- These computer-based simulations enable architects and engineers to evaluate the performance of their designs.



- Dimensional simulation is a technique used to analyze and visualize the behavior of complex systems across different dimensions or parameters.
- It involves varying one or more dimensions of a model to observe how changes affect the overall system's performance, outcomes, or behaviors.
- Involves creating models that can operate in multiple dimensions, such as spatial dimensions (2D or 3D) or parametric dimensions (different variables affecting the system).
- Systematic exploration of how changes in specific parameters (like temperature, pressure, or material properties) impact the system. This helps identify optimal conditions or critical thresholds.

- Visual tools to represent data across multiple dimensions, making it easier to interpret complex interactions and results like graphs, plots etc
- Examples are designing the angle of an aircraft wing, heating of a metal plate etc.

Types of Mathematical Simulations:

1. **Deterministic Simulations:** These involve systems where the outcomes are completely determined by initial conditions and model parameters.
2. **Stochastic Simulations:** These include randomness or uncertainty, often used for probabilistic systems.

Advantages

Realistic representation, can test real world constraints.

Disadvantages

Expensive, time-consuming, limited to physical scales.

Applications

Aerodynamics, structural engineering, fluid dynamics.

1.2.8.2 VIRTUAL SIMULATION

Virtual simulation refers to the use of computer-generated environments to model real-world processes, objects, or systems, allowing users to interact with a simulated reality. These simulations replicate the behavior and experience of actual systems, often in real-time, and are widely used in training, education, gaming, and scientific research.

VR Surgery Simulators train doctors for delicate operations, offering a practice environment for tasks like heart surgery or laparoscopic procedures.

Flight Simulators like those used by NASA or airlines simulate turbulence, engine failure, and bad weather to prepare pilots for various flight scenarios.

Virtual Battlefield Simulators like VBS3 allow soldiers to practice tactics, teamwork, and decision-making in a controlled virtual space.

1.2.8.3 Physical Simulation:

A physical simulation uses computational models to replicate the physical behavior of systems or objects under various conditions based on real-world physics.

Characteristics:

- **Real-World Physics:** Uses established laws of physics to model real-world processes.
- **Computational Modeling:** Involves extensive calculations and data to simulate behaviors like motion, deformation, heat transfer, or fluid flow.
- **Visual Representation:** Often results in a 3D or visual output that mimics physical behaviors

A **scale-down model** is a smaller, simplified version of a real-world system or object designed to replicate the behavior or functionality of the larger system.

These models retain the essential features of the original system but are scaled down in size, cost, or complexity for practical purposes.



